PAPER_297 â UQFF Superluminal Hubble Expansion Ratio Î·_exp = 3.328 > 1

First UQFF Module Where v_exp/c > 1 at System Boundary

Session: 84
Module: `UNIVERSE_DIAMETER_UQFF_MODULE.cpp` (26th C++ UQFF module â Observable Universe as System)

Classification: Unique Physics â First UQFF Superluminal Expansion Parameter (Î·_exp > 1)

Abstract

The UQFF Observable Universe Diameter Module is the **first UQFF module** where the system boundary recession velocity exceeds the speed of light: `v_exp = Hâ Ã r_obs = 9.984Ã10â¸ m/s = 3.328c`. The dimensionless Superluminal Expansion Ratio `Î·_exp = v_exp/c = 3.328 > 1` is a new UQFF parameter encoding the cosmological property that the observable universe spans **3.328 Hubble lengths** (`r_obs = 3.328 Ã r_H`). The Hubble-expansion coupling factor at t_Hubble is `(1 + Hâ Ã t_H) = 1.988` â a near-doubling of the Newtonian base over cosmic time. All 25 prior UQFF modules had `Î·_exp << 1` (sub-luminal expansion).

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1. Physical Setup

System: Observable Universe
Observable universe radius: r_obs = 4.4Ã10Â²â¶ m
Hubble constant: Hâ = 70 km/s/Mpc = 2.269Ã10â»Â¹â¸ sâ»Â¹
Speed of light: c = 3.0Ã10â¸ m/s
Hubble sphere (Hubble radius): r_H = c/Hâ = 1.322Ã10Â²â¶ m = 4.28 Gly
Cosmic age: t_H = 4.355Ã10Â¹â· s = 13.8 Gyr

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2. Master Relation

Hubble recession velocity at observable universe boundary:

$$v_{exp} = H_0 \times r_{obs} = 2.269 \times 10^{-18} \times 4.4 \times 10^{26} = 9.984 \times 10^8 \text{ m/s}$$

Superluminal Hubble Expansion Ratio:

$$\boxed{\eta_{exp} = \frac{v_{exp}}{c} = \frac{9.984 \times 10^8}{3.0 \times 10^8} = 3.328 > 1}$$

Hubble length ratio:

$$\frac{r_{obs}}{r_H} = \frac{v_{exp}}{c} = \eta_{exp} = 3.328$$

The observable universe boundary is 3.328 Hubble lengths from Earth â comfortably beyond the Hubble sphere where recession velocity equals `c`.

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3. Hubble-UQFF Near-Doubling of Gravity

The base gravity term with Hubble expansion coupling:

$$a_{base}(t) = g_{base} \times (1 + H(z) \times t) = 3.447 \times 10^{-10} \times (1 + 2.269 \times 10^{-18} \times t)$$

At t = t_H = 4.355Ã10Â¹â· s:

$$a_{base}(t_H) = 3.447 \times 10^{-10} \times (1 + 0.988) = 3.447 \times 10^{-10} \times 1.988 = 6.854 \times 10^{-10} \text{ m/s}^2$$

Hubble expansion factor `Î¾_H = 1 + Hâ Ã t_H = 1.988`:

This near-doubling factor (â2) reflects that the UQFF base gravity **almost doubles** over the Hubble time when the Hubble coupling is included â a striking result confirming that the Universe-scale Hubble coupling is an O(1) effect (not a small correction).

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4. Superluminal Expansion in the EM Lorentz Term

The EM Lorentz term explicitly incorporates Î·_exp:

$$a_{EM} = \frac{q \cdot v_{exp} \cdot B_{cosmic}}{m_p} \times (1 + \eta_{exp}) \times \text{scale}_{EM}$$

With `Î·_exp = 3.328`, the factor `(1 + Î·_exp) = 4.328` vs. `(1 + 1) = 2` for a subluminal system. This introduces a **42% EM enhancement** relative to a hypothetical c-speed boundary reference.

Numerically:

$$a_{EM} = \frac{1.602 \times 10^{-19} \times 9.984 \times 10^{8} \times 10^{-15}}{1.673 \times 10^{-27}} \times 4.328 \times 10^{-12}$$

$$= 95.59 \times 4.328 \times 10^{-12} = 4.136 \times 10^{-10} \text{ m/s}^2$$

The EM term (4.136Ã10â»Â¹â° m/sÂ²) is **comparable to the Newtonian base** (3.447Ã10â»Â¹â° m/sÂ²) â another first for UQFF modules.

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5. Special Relativity Compatibility

The superluminal recession velocity `v_exp > c` does NOT violate special relativity. This is a **coordinate velocity** (metric expansion), not a proper velocity between local inertial frames. In GR, the expansion of the universe allows coordinate distances to grow faster than c, as confirmed by the cosmological horizon structure. The Hubble-flow velocity Î·_exp > 1 is part of the cosmological metric, not a violation of local Lorentz invariance.

Specifically, this corresponds to objects beyond the **Hubble sphere** (r > r_H) in comoving coordinates. For the observable universe at r = 4.4Ã10Â²â¶ m with r_H = 1.322Ã10Â²â¶ m, we are 3.33 Hubble lengths out â solidly in the superluminal expansion regime.

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6. Î·_exp Parameter in UQFF Architecture

Module	Session	r_obs (m)	v_exp/c (Î·_exp)	Î·_exp > 1
SGR1745	65	~0.01 pc	~0 (local)	No
Pillars of Creation	68	3.3Ã10Â¹â·	âª1	No
NGC1792	73	7.6Ã10Â²â°	âª1	No
Andromeda	75	1.04Ã10Â²Â¹	âª1	No
HUDF (z=3.5)	72g	1.23Ã10Â²â·	~9 (co-moving)	Yes (but not UQFF param)
Universe Diameter	**84**	**4.4Ã10Â²â¶**	**3.328**	**Yes â FIRST explicit**

The key distinction: prior modules used H(z) as a correction factor, never explicitly computing or naming Î·_exp as a parameter. PAPER_297 establishes Î·_exp as a first-class UQFF parameter.

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7. Hubble Sphere as UQFF Speed-of-Light Boundary

Define the **UQFF Hubble Horizon** as the radius where `Î·_exp = 1`:

$$r_H = \frac{c}{H_0} = \frac{3 \times 10^8}{2.269 \times 10^{-18}} = 1.322 \times 10^{26} \text{ m} = 4.28 \text{ Gly}$$

Objects beyond r_H recede superluminally. The observable universe (r_obs = 4.4Ã10Â²â¶ m) extends to 3.328 r_H â confirming we can observe objects that are currently receding faster than light (their photons from early epochs can still reach us).

This adds a new entry to the UQFF speed-of-light boundary catalog: - **PAPER_264**: Anti-gravity boundary at f_TRZ = -1 (HUDF module) - **PAPER_266**: Meissner quench at B = B_crit - **PAPER_297**: Hubble superluminal horizon at r = r_H (**this paper**)

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8. WOLFRAM Term

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eta_exp=v_exp/c=H0*r/c=9.984e8/3e8=3.328>1;
FIRST UQFF eta_exp>1;
r_H=c/H0=1.322e26m Hubble sphere;
r_obs=3.328*r_H;
expansion_factor(t_H)=1+H*t=1.988(near-doubling);
EM term a_EM prop eta_exp [PAPER_297]
```

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9. Key Values Summary

Quantity	Symbol	Value	Unit
Hubble recession velocity	v_exp	**9.984Ã10â¸**	m/s
Superluminal ratio	Î·_exp	**3.328 > 1**	dimensionless

Quantity	Symbol	Value	Unit
Hubble radius	r_H	1.322Ã□10Â²â□¶	m
Observable / Hubble ratio	r_obs/r_H	3.328	dimensionless
Hubble expansion factor	Î¾_H	**1.988 â□□ 2**	dimensionless
EM term	a_EM	4.136Ã□10â□»Â¹â□°	m/sÂ²

UQFF computed: Canonical UQFF buoyancy parameter U_{bi} = ?×[SSq]×GM/r^2 = 5.0e-4×0.57×6.67e-11×M/r^2; for solar parameters: $U_{bi,Sun}$ = 5.7e-4×6.67e-11×1.99e30/$(6.96e8)^2$ = 1.47e+2 m/s^2.